

Rolling of a Homogeneous Ball on a Moving Cylinder

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Abstract—This paper addresses the problem of a homogeneous ball rolling on the inner surface of a circular cylinder in a field of gravity parallel to its axis. It is assumed that the ball rolls without slipping on the surface of the cylinder, and that the cylinder executes plane-parallel motions in a circle perpendicular to its symmetry axis. The integrability of the problem by quadratures is proved. It is shown that in this problem the trajectories of the ball are quasi-periodic in the general case, and that an unbounded elevation of the ball is impossible. However, in contrast to a fixed (or rotating) cylinder, there exist resonances at which the ball moves on average downward with constant acceleration.

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1. INTRODUCTION

As is well known, the problem of the nonholonomic rolling of a homogeneous ball on the surface of a vertical cylinder is a good illustration of the unrealistic nature of some conclusions which can be drawn using nonholonomic mechanics. In particular, the results of investigation of this problem suggest that under almost all initial conditions (except for straight-line rolling motion downwards) the ball will not move downward, but will execute periodic or quasi-periodic rotations in a vertically bounded region. This is a consequence of the integrability of the problem of a ball rolling on a cylinder, which was proved by Routh in 1892 [1]. Moreover, he showed that the problem of a homogeneous ball rolling without slipping on an arbitrary surface of rotation is integrable.

The problem of a ball rolling on the surface of a cylinder of an arbitrary cross-section is a generalization of the above problem. The integrability of this problem by quadratures was first shown by Stübler [2] in 1909. In addition, in [2] it was shown that there is no drift of the ball along the cylinder. In the modern literature, an analysis of quadratures of quasi-periodic functions similar to those arising in [2] can be found, for example, in [3]. In [4] the Liouville integrability of this problem is proved and the case of an inclined cylinder is considered.

Further generalizations of the problem of a ball rolling on a cylinder involved the rotation of the supporting cylinder and incorporation of friction forces and friction torques. The integrability of the problem of a ball rolling on a rotating cylinder was also shown by Routh [1]. An analysis of quadratures in the problem of a ball rolling on a circular cylinder in the presence of viscous friction was made in [5], and a qualitative analysis of the problem of a ball rolling on a rotating cylinder,

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in particular, in the presence of friction was carried out in [6]. The dynamics of a ball on various surfaces of rotation, including rotating ones, was explored in [7–14].

Note that, despite the “unphysicality” of the nonholonomic model (due to the absence of dissipation in it), it provides a fairly good description of some observable phenomena, in particular, the phenomena arising at times (or at velocities) at which dissipation can be neglected. The simplest example is this: a basketball thrown into the basket can execute several rotations along the rim and then fly back without getting into the basket. There has been a large amount of research on such behavior of a basketball [15, 16]. Another example that illustrates the dynamics of a ball in a vertical cylinder is the well-known phenomenon of a golf ball popping out of the hole. A physical explanation of this phenomenon and the results of a full-scale experiment can be found in [17]. A video of such experiments can be found at <https://www.youtube.com/watch?v=5sbM2Isx17A>.

In this paper, we address another problem similar to those described above: the rolling motion of a ball on the surface of a cylinder executing plane-parallel circular motions. This problem is concerned with a classical phenomenon observed in experiment, which can be called a “ball in a glass”. As is well known, when performing circular motions of a glass, it is easy to make the ball pop out of it. In this paper, we construct a model of the motion of a ball on the surface of a moving cylinder and make an attempt at explaining the observed phenomenon using the nonholonomic model.

2. EQUATIONS OF MOTION

Consider the rolling motion of a homogeneous ball of mass m and radius a on the inner surface of a vertical circular cylinder of radius R in a field of gravity parallel to the axis of the cylinder (see Fig. 1). It is assumed that the cylinder itself executes periodic plane-parallel motions in a circle whose plane is perpendicular to the cylinder’s axis. That is, the cylinder’s axis traces out a circle of radius ρ . It is also assumed that the ball rolls on the cylinder’s surface without slipping (the velocity of the contact point relative to cylinder is zero) and has only one point, P , of contact with the supporting surface.

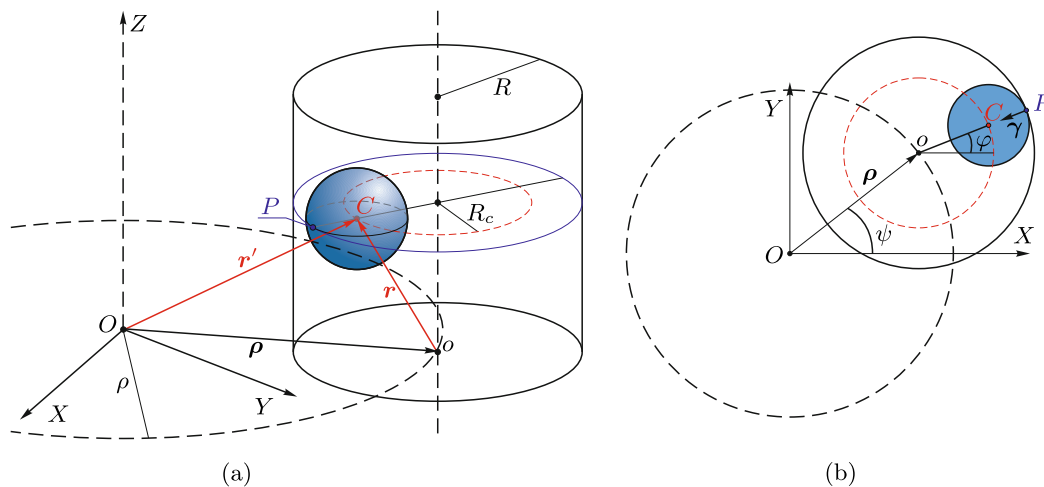


Fig. 1. (a) Diagrammatic representation of the system under consideration, (b) top view.

To describe the motion of the ball, we introduce a fixed (inertial) coordinate system $OXYZ$ with axis OZ directed vertically upward and with origin at the center of the circle along which the cylinder’s axis moves.

The position of the axis of the moving cylinder relative to the fixed coordinate system $OXYZ$ will be given by the vector

$$\rho = (\rho \cos \Omega t, \rho \sin \Omega t, 0),$$

where $\Omega = \dot{\psi} = \text{const}$ is the angular velocity of rotation of the cylinder and ψ is the angle between the axis OX and the vector $\boldsymbol{\rho}$ (it defines the position of the cylinder, see Fig. 1).

Since there is no slipping, the velocity of the contact point P on the ball is the same as the velocity of the point on the rotating surface which comes in contact with it, i. e.,

$$\dot{\mathbf{r}}' + a\boldsymbol{\gamma} \times \boldsymbol{\omega} = \dot{\boldsymbol{\rho}}, \quad (2.1)$$

where $\mathbf{r}' = \mathbf{r} + \boldsymbol{\rho}$ is the radius vector of the center of mass of the ball, $\mathbf{r}$ is the radius vector of the center of mass of the ball drawn from point o (see Fig. 1), $\boldsymbol{\omega}$ is the angular velocity of the ball, and $\boldsymbol{\gamma}$ is the normal vector to the surface of the cylinder at the point of contact of the ball with the surface of the cylinder.

The normal vector to the supporting surface at the point of contact of the ball with the cylinder $\boldsymbol{\gamma}$ has the form

$$\boldsymbol{\gamma} = -\frac{1}{R_c} \mathbf{k} \times (\mathbf{r} \times \mathbf{k}),$$

where $R_c = R - a$ is the radius of the cylinder on which the center of mass of the ball moves, and $\mathbf{k} = (0, 0, -1)$ is the vector directed vertically downward.

We write the change in the linear and angular momentum of the ball in the form of Newton–Euler equations

$$\begin{aligned} m\ddot{\mathbf{r}}' &= \mathbf{N} + \mathbf{F}, \\ I\dot{\boldsymbol{\omega}} &= a\mathbf{N} \times \boldsymbol{\gamma} + \mathbf{M}_F, \end{aligned} \quad (2.2)$$

where $\mathbf{N}$ is the reaction of the supporting cylinder at the point of contact with the ball, $\mathbf{F}$ is the resultant of external active forces applied at the center of mass of the ball, and $\mathbf{M}_F$ is the resultant moment of external forces relative to the center of mass of the ball.

In what follows we will assume that only gravity force acts on the system, and therefore the force has the form $\mathbf{F} = mg\mathbf{k}$. Then the resultant moment induced by this force will be equal to zero: $\mathbf{M}_F = 0$.

Expressing the reaction from the first equation of (2.2) and substituting into the second, we obtain

$$I\dot{\boldsymbol{\omega}} = ma(\ddot{\mathbf{r}}' - g\mathbf{k}) \times \boldsymbol{\gamma}. \quad (2.3)$$

Let us eliminate the second time derivative $\ddot{\mathbf{r}}'$ using the derivative of Eq. (2.1). Then Eq. (2.3) takes the form

$$\mathbf{J}\dot{\boldsymbol{\omega}} = ma(\ddot{\boldsymbol{\rho}} - a\dot{\boldsymbol{\gamma}} \times \boldsymbol{\omega} - g\mathbf{k}) \times \boldsymbol{\gamma},$$

where we have introduced the notation $\mathbf{J}\dot{\boldsymbol{\omega}} = I\dot{\boldsymbol{\omega}} + ma^2\boldsymbol{\gamma} \times (\dot{\boldsymbol{\omega}} \times \boldsymbol{\gamma})$.

To close the system, we supplement the resulting equation with relation (2.1) written in terms of the radius vector $\mathbf{r}$. This yields

$$\begin{aligned} \mathbf{J}\dot{\boldsymbol{\omega}} &= ma(\ddot{\boldsymbol{\rho}} - a\dot{\boldsymbol{\gamma}} \times \boldsymbol{\omega} - g\mathbf{k}) \times \boldsymbol{\gamma}, \\ \dot{\mathbf{r}} + a\boldsymbol{\gamma} \times \boldsymbol{\omega} &= 0. \end{aligned} \quad (2.4)$$

The resulting equations of motion (2.4) admit the obvious geometric integral of motion

$$\mathcal{F}_0 = r_1^2 + r_2^2 = R_c^2$$

and two additional integrals. The first is the part of energy that does not depend on the vertical component of the angular velocity

$$\mathcal{F}_1 = \frac{1}{2}(I + ma^2)(\boldsymbol{\omega}, \boldsymbol{\gamma} \times \mathbf{k})^2 + \frac{1}{2}I(\boldsymbol{\omega}, \boldsymbol{\gamma})^2 - mg(\mathbf{r}, \mathbf{k}).$$

The second integral is nonautonomous and corresponds to the “moving” energy of horizontal motion relative to some moving coordinate system

$$\mathcal{F}_2 = \frac{1}{2}(I + ma^2)\left(\omega_3 + \frac{R_c}{a}\Omega\right)^2 + m(\mathbf{r}, \ddot{\boldsymbol{\rho}}).$$

Remark. In the case of a fixed cylinder the integral $\mathcal{F}_2$ reduces to $\omega_3 = \text{const}$, and the sum of integrals $\mathcal{F}_1 + \mathcal{F}_2$ reduces to the total energy of the system.

In addition, the system (2.4) possesses a symmetry field corresponding to a shift along the vertical

$$\zeta = \frac{\partial}{\partial r_3}$$

and has the standard invariant measure $d\omega_1 d\omega_2 d\omega_3 dr_1 dr_2 dr_3$.

To simplify the form of the equations and analyze the resulting system, we introduce new variables¹⁾

$$\begin{aligned} \omega_\gamma = (\boldsymbol{\omega}, \boldsymbol{\gamma}) &= -\frac{\omega_1 r_1 + \omega_2 r_2}{R_c}, \quad \omega_\varphi = (\boldsymbol{\omega}, \boldsymbol{\gamma} \times \mathbf{k}) = \frac{\omega_1 r_2 - \omega_2 r_1}{R_c}, \\ \omega_z &= \omega_3, \quad z = r_3, \quad \varphi = \arctan(r_2, r_1). \end{aligned} \quad (2.5)$$

The initial variables can be obtained by making the inverse transformation

$$\begin{aligned} r_1 &= R_c \cos \varphi, \quad r_2 = R_c \sin \varphi, \quad r_3 = z, \\ \omega_1 &= \omega_\varphi \sin \varphi - \omega_\gamma \cos \varphi, \quad \omega_2 = -(\omega_\varphi \cos \varphi + \omega_\gamma \sin \varphi), \quad \omega_3 = \omega_z. \end{aligned}$$

In the new variables (2.5) the equations of motion (2.4) become

$$\begin{aligned} \dot{\omega}_\gamma &= -\frac{a}{R_c} \omega_\varphi \omega_z, \\ \dot{\omega}_\varphi &= \frac{a}{R_c(I + ma^2)} (mgR_c + I\omega_\gamma \omega_z), \\ \dot{\omega}_z &= \frac{ma\rho\Omega^2}{I + ma^2} \sin(\varphi - \psi(t)), \\ \dot{\varphi} &= -\frac{a}{R_c} \omega_z, \\ \dot{z} &= -a\omega_\varphi. \end{aligned} \quad (2.6)$$

We note that Eqs. (2.6) decouple into the reduced system for the variables φ , ω_φ , ω_γ , ω_z and the quadrature for the coordinate z .

We transform the resulting equations to dimensionless form. To do so, we choose the ball's mass m as a unit of mass, R_c as a unit of length, and the expression $\sqrt{R_c/g}$ as a unit of time. This is equivalent to the relations

$$m = 1, \quad R_c = 1, \quad g = 1.$$

Next, we introduce new parameters

$$\chi^2 = \frac{I}{I + a^2}, \quad \omega_0^2 = (1 - \chi^2)\rho\Omega^2, \quad \mu^2 = \frac{1 - \chi^2}{I}$$

and make the scaling transformation

$$\omega_z = \frac{\tilde{\omega}_z}{a}, \quad \omega_\varphi = \chi \tilde{\omega}_\varphi.$$

¹⁾The function $\arctan(\cdot, \cdot)$ with two arguments calculates the arctangent of the quotient of its arguments and takes into account their signs in determining the value of the function in the interval $[0, 2\pi)$. In some cases it is more convenient to use the formula

$$\varphi = 2 \arctan \left(\frac{r_2}{R_c + r_1} \right),$$

because the tangent of the half-angle is uniquely defined.

As a result, the reduced system takes the form (the signs $\sim$ are omitted)

$$\begin{aligned}\dot{\omega}_\gamma &= -\chi\omega_\varphi\omega_z, \\ \dot{\omega}_\varphi &= \chi\omega_\gamma\omega_z + \mu, \\ \dot{\omega}_z &= \omega_0^2 \sin(\varphi - \Omega t), \\ \dot{\varphi} &= -\omega_z,\end{aligned}\tag{2.7}$$

and the quadrature for the coordinate z is

$$\dot{z} = -a\chi\omega_\varphi.\tag{2.8}$$

In the new variables the first integral $\mathcal{F}_1$ takes the form

$$\mathcal{F}_1 = \frac{1}{2}(\omega_\varphi^2 + \omega_\gamma^2) + \frac{\mu}{a\chi}z.\tag{2.9}$$

This integral can be used instead of the quadrature (2.8) to define the dependence $z(t)$ after solving the reduced system (2.7).

In the new variables the integral $\mathcal{F}_2$ of the complete system takes the form

$$\mathcal{F}_2 = \frac{1}{2}(\omega_z + \Omega)^2 - \omega_0^2 \cos(\varphi - \Omega t)\tag{2.10}$$

and is a (nonautonomous) first integral of the reduced system.

In addition, the reduced system admits the standard invariant measure $d\omega_\gamma d\omega_\varphi d\omega_z d\varphi$.

Remark. By making the transformation $\varphi - \Omega t = \tilde{\varphi}$ the system (2.7) and the integral (2.10) can be reduced to autonomous form. However, we will not do that because the solution of the reduced system (2.7) in terms of quadratures can be obtained without such a transformation as well. We only note that, after such a transformation, the integral (2.10) is obviously an analog of the “moving” energy or is a Jacobi integral (see [8, 10]).

It is easy to see that the reduced system (2.7) decouples into two subsystems. The first subsystem — for the variables (ω_z, φ) — is the system of equations of a mathematical pendulum. The second subsystem — for $(\omega_\gamma, \omega_\varphi)$ — is the system of equations of an oscillator with time-dependent frequency and a constant external force. Below we present the solution of the system (2.7) in terms of quadratures.

3. INTEGRATION BY QUADRATURES

We first consider the subsystem for (ω_z, φ)

$$\begin{aligned}\dot{\omega}_z &= \omega_0^2 \sin(\varphi - \Omega t), \\ \dot{\varphi} &= -\omega_z.\end{aligned}\tag{3.1}$$

Equations (3.1) admit the integral $\mathcal{F}_2$ and, as is well known (see, e.g., [18, 19] or other textbooks on theoretical mechanics), can be integrated in terms of elliptic functions. Depending on the level set of the integral $\mathcal{F}_2 = h$, the solutions of the system (3.1) have the following form.

1° If $h = -\omega_0^2$, which corresponds to the minimum of the integral $\mathcal{F}_2$, we find that the solution of the system (3.1) has the form

$$\varphi = \Omega t, \quad \omega_z = -\Omega.$$

2° If $|h| < \omega_0^2$, then the solution of the system of equations is expressed in terms of the elliptic functions

$$\varphi = \Omega t - 2 \arcsin(k_1 \operatorname{sn}(\omega_0 t - \tau_1, k_1)), \quad \omega_z = 2\omega_0 k_1 \operatorname{cn}(\omega_0 t - \tau_1, k_1) - \Omega,\tag{3.2}$$

where

$$k_1 = \frac{\sqrt{\frac{1}{2}(h + \omega_0^2)}}{\omega_0},$$

the constant of integration τ_1 is related to the initial angle φ_0 by $\tau_1 = F\left(\arcsin\left(\frac{1}{k_1}\sin\frac{\varphi_0}{2}\right), k_1\right)$, and the period of the elliptic functions $\operatorname{sn}(\omega_0 t - \tau_1, k_1)$ and $\operatorname{cn}(\omega_0 t - \tau_1, k_1)$ is

$$T_1 = \frac{4K(k_1)}{\omega_0}, \quad (3.3)$$

where $K(k_1) = F(\pi/2, k_1)$ is the total elliptic integral of the first kind.

3° The case $h = \omega_0^2$ corresponds to two solutions of the system (3.1). The first solution corresponds to the local maximum of the integral $\mathcal{F}_2$

$$\varphi = \Omega t - \pi, \quad \omega_z = -\Omega.$$

The second solution corresponds to asymptotic motions of a pendulum and has the form

$$\varphi = \Omega t + \pi - 4 \arctan(e^{\pm \omega_0 t}), \quad \omega_z = \pm \frac{2\omega_0}{\cosh \omega_0 t} - \Omega. \quad (3.4)$$

4° If $|h| > \omega_0^2$, then the solution of the system (3.1) is also represented by elliptic functions and has the form

$$\varphi = \Omega t - 2 \operatorname{am}(k_1 \omega_0 t - \tau_2, k_2), \quad \omega_z = 2k_1 \omega_0 \operatorname{dn}(k_1 \omega_0 t - \tau_2, k_2) - \Omega,$$

where $k_2 = \frac{1}{k_1}$. The constant of integration τ_2 is related to the initial angle φ_0 by $\tau_2 = F\left(\frac{\varphi_0}{2}, k_2\right)$. The period of the elliptic function $\operatorname{dn}(k_1 \omega_0 t - \tau_2, k_2)$ is

$$T_2 = \frac{2K(k_2)k_2}{\omega_0}. \quad (3.5)$$

Further, knowing $\omega_z(t)$ from the solution of the subsystem (3.1), we obtain the solution of the subsystem for the variables $(\omega_\gamma, \omega_\varphi)$

$$\begin{aligned} \dot{\omega}_\gamma &= -\chi \omega_z \omega_\varphi, \\ \dot{\omega}_\varphi &= \chi \omega_z \omega_\gamma + \mu. \end{aligned} \quad (3.6)$$

To solve this system, we introduce the complex variable $Z = \omega_\varphi + i\omega_\gamma$, where i is a complex unit. The system (3.6) for the variable Z reduces to one equation, which, using the second equation of (3.1), takes the form

$$\dot{Z} = i\chi\dot{\varphi}Z + \mu.$$

The general solution of this equation can be written as

$$Z = \left(\mu \int e^{-i\chi\varphi} dt + C_0\right) e^{i\chi\varphi}, \quad (3.7)$$

where $C_0 = a_0 + ib_0$ is the constant of integration. The angular velocities ω_φ and ω_γ are defined as the real and the imaginary part of Z

$$\omega_\varphi = \operatorname{Re}Z(t), \quad \omega_\gamma = \operatorname{Im}Z(t). \quad (3.8)$$

Below we analyze the quadratures obtained above from the viewpoint of the dynamical behavior of the ball in absolute space.

4. ANALYSIS OF DYNAMICS

4.1. Boundedness of Motion

To estimate the boundedness (above or below) of the ball's motion, it is necessary to determine what values the variable z can take. From the expression for the integral $\mathcal{F}_1$

$$\mathcal{F}_1 = \frac{1}{2}(\omega_\varphi^2 + \omega_\gamma^2) + \frac{\mu}{a\chi}z$$

it is evident that on the fixed level set of the integral $\mathcal{F}_1 = \mathcal{F}_1^0 = \text{const}$ the value of the coordinate z is bounded above by the value

$$z_{\max} = \frac{a\chi}{\mu}\mathcal{F}_1^0, \quad (4.1)$$

and the value of z is not bounded below. Moreover, from the form of the integral $\mathcal{F}_1$ one can conclude that the lower the ball descends, the higher become its angular velocities ω_γ and/or ω_φ .

Proposition. *In the problem of a ball rolling on the surface of a cylinder executing circular motions, the vertical coordinate of the ball is bounded above by the value (4.1).*

4.2. General Motion

We now consider the question of the motion pattern of the system in the general case. For this we analyze the quadratures obtained in the preceding section for different values of the integral $\mathcal{F}_2 = h$.

1° For $h = -\omega_0^2$ the solution of the reduced system can be written as

$$\omega_\gamma = \frac{\mu}{\chi\Omega} + H \sin(\chi\Omega t + \Phi_0), \quad \omega_\varphi = H \cos(\chi\Omega t + \Phi_0), \quad \omega_z = -\Omega, \quad \varphi = \Omega t, \quad (4.2)$$

where H and Φ_0 are constants of integration which are defined from the initial conditions. The coordinate z changes periodically:

$$z = -\frac{aH}{\Omega} \sin(\chi\Omega t + \Phi_0). \quad (4.3)$$

Here, by virtue of the translation symmetry, the constant of integration of the quadrature (2.8) is omitted.

From (4.2) and (4.3) it follows that if the parameter χ is a rational number, then the trajectory of the ball is periodic. Otherwise the trajectory of the ball is a quasi-periodic winding of a segment of the cylinder. In both cases the trajectory of the ball is bounded in the vertical and lies in the strip

$$|z| < \frac{aH}{\Omega}. \quad (4.4)$$

A special case of these motions with $H = 0$ is the solution

$$\omega_\gamma = \frac{\mu}{\chi\Omega}, \quad \omega_\varphi = 0, \quad \omega_z = -\Omega, \quad \varphi = \Omega t, \quad (4.5)$$

which is a motion during which the ball is always at the maximal distance $\rho + R_c$ from the center of rotation of the cylinder (see Fig. 2). The vertical coordinate of the center of the ball does not change: $z = \text{const}$.

We analyze the linear orbital stability of this solution. To do so, we introduce small perturbations of the solution (4.5) and obtain a linear system describing the changes in these perturbations. The characteristic equation for this linear system has the form

$$P_4(\lambda) = (\Omega^2\chi^2 + \lambda^2)(\omega_0^2 + \lambda^2),$$

where λ are the roots of the characteristic equation. In this case all roots are purely imaginary, and hence the solution (4.5) is stable.

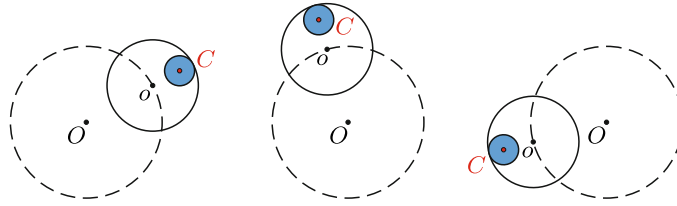


Fig. 2. Positions of the ball and the cylinder for the case of solution (4.5) at different time instants (top view).

2° For $|h| < \omega_0^2$ the solution of the reduced system (2.7) is given by Eqs. (3.7) and (3.2). Let us represent the dependence $\varphi(t)$ as

$$\varphi = \Omega t + \varphi_0(t),$$

where $\varphi_0(t)$ is a periodic function with period T_1 (3.3). Next, we define the dependence of the coordinate $z(t)$. To do so, we represent the integrand in (3.7) as a Fourier series

$$e^{-i\chi\varphi(t)} = e^{-i\chi\Omega t} \sum_{n=-\infty}^{\infty} d_n e^{in\omega_1 t}, \quad (4.6)$$

where d_n are the coefficients of expansion of the function $e^{-i\chi\varphi_0(t)}$, and

$$\omega_1 = \frac{2\pi}{T_1} = \frac{\pi\omega_0}{2K(k_1)}.$$

This can always be done since $\varphi_0(t)$ is a periodic function.

Substituting (4.6) into (3.7), we obtain

$$Z = \left(\mu \sum_{n=-\infty}^{\infty} \frac{d_n}{-i(\chi\Omega - n\omega_1)} e^{-i(\chi\Omega - n\omega_1)t} + C_0 \right) e^{i\chi\varphi(t)}.$$

Analyzing the resulting expression, we can single out two qualitatively different cases of the system's behavior.

- 1) Nonresonant case. In this case, the relation $\chi\Omega - n\omega_1 \neq 0$ is satisfied for $\forall n \in \mathbb{Z}$. In the general case where the ratio of the frequencies is incommensurable, i. e., $\frac{\chi\Omega}{\omega_1} \notin \mathbb{Q}$, the angular velocities ω_γ and ω_φ change quasi-periodically. If the frequencies are commensurable, $\frac{\chi\Omega}{\omega_1} \in \mathbb{Q}$, then these angular velocities are periodic functions of time. In both cases, expressing z from the integral $\mathcal{F}_1$, we find that

$$z = \left(z_{\max} - \frac{a\chi}{2\mu} (\omega_\gamma^2 + \omega_\varphi^2) \right). \quad (4.7)$$

This implies that the coordinate z is also a periodic or quasi-periodic function, and hence it is bounded both above and below.

- 2) Resonant case. In this case there exists an $n \in \mathbb{Z}$ such that the resonant relation $\chi\Omega - n\omega_1 = 0$ is satisfied. Then $Z(t)$ can be represented as

$$Z(t) = t\Psi^*(t) + \Psi_0(t) + \Psi_1(t),$$

where $\Psi^*(t)$ and $\Psi_1(t)$ are quasi-periodic functions and $\Psi_0(t)$ is a periodic function with period T_1 . From this expression, as well as from (4.7), it follows that the coordinate z decreases on average quadratically with time.

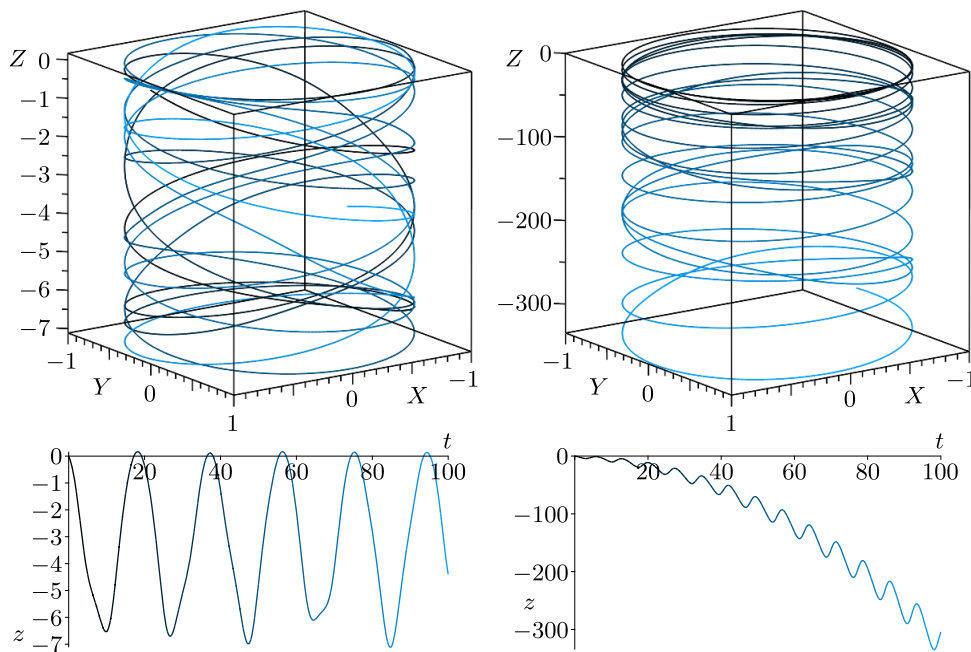


Fig. 3. An example of the trajectory of the center of the ball and function $z(t)$ in the general case (left) and in the resonant case (right). The graphs were obtained with the same initial conditions and the following values of parameters: $\mu = 1.13$, $\omega_0 = 0.88$, $\Omega = 1.1$, $a = 1$ and $\chi = 0.3$ (left), $\chi = \frac{\omega_0}{\Omega} \approx 0.776$ (right).

Figure 3 shows an example of trajectories of the center of the ball for the general case and the resonant case, and the corresponding dependences of the coordinate z on time. For visual clarity, the color of the curves in the figures depends on time and changes from black (at $t = 0$) to light-blue (at $t = 100$).

3° On the level set $h = \omega_0^2$ there are two different solutions of the reduced system. The first is a solution similar to (4.2):

$$\omega_\gamma = \frac{\mu}{\chi\Omega} + H \sin(\chi\Omega t + \Phi_0), \quad \omega_\varphi = H \cos(\chi\Omega t + \Phi_0), \quad \omega_z = -\Omega, \quad \varphi = \Omega t + \pi. \quad (4.8)$$

As in the case (4.2), the solution (4.8) is periodic in the case $\chi \in \mathbb{Q}$ and quasi-periodic if this is not the case. The coordinate z also changes periodically (4.3) and lies in the strip (4.4).

The particular case of the solution (4.8) with $H = 0$ has the form

$$\omega_\gamma = \frac{\mu}{\chi\Omega}, \quad \omega_\varphi = 0, \quad \omega_z = -\Omega, \quad \varphi = \Omega t + \pi, \quad (4.9)$$

and corresponds to a rolling motion of the ball on the cylinder's surface such that the ball is always at a minimal distance $\rho - R_c$ from the center of rotation of the cylinder (see Fig. 4). The coordinate z on this solution does not change: $z = \text{const}$.

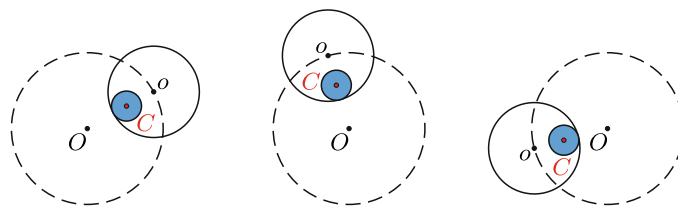


Fig. 4. Positions of the ball and the cylinder for the case of solution (4.9) at different time instants (top view).

Analysis of the orbital stability of the solution (4.9) leads to the characteristic equation

$$P_4(\lambda) = (\Omega^2 \chi^2 + \lambda^2)(\omega_0^2 - \lambda^2).$$

Two roots of the equation are purely imaginary, and the other two are real roots with opposite signs. Hence, this solution is unstable.

The second solution is defined by Eqs. (3.7), (3.4) and is asymptotic. As $t \rightarrow \pm\infty$ this solution tends to the solutions (4.8). Thus, the trajectory of the ball in this case is bounded.

4° For $|h| > \omega_0^2$ the dependence $\varphi(t)$ can be represented as

$$\varphi = (\Omega - \omega_2)t + \tilde{\varphi}_0(t),$$

where $\tilde{\varphi}_0(t)$ is a periodic function with period T_2 , and

$$\omega_2 = \frac{2\pi}{T_2} = \frac{\pi\omega_0}{K(k_2)k_2}.$$

Proceeding as in 2°, one can show that in the general case the motion is periodic or quasi-periodic, and the coordinate z is bounded. In the resonant case, if the relation $\chi(\Omega - \omega_2) - n\omega_2 = 0$ is satisfied for some $n \in \mathbb{Z}$, the coordinate z decreases on average quadratically in time.

Based on the results presented above, we can formulate the following proposition:

Proposition. *The trajectories of a homogeneous ball rolling without slipping on the surface of a vertical cylinder executing plane-parallel horizontal circular motions*

- 1) *are bounded above under any initial conditions;*
- 2) *are bounded below under almost all initial conditions.*

CONCLUSION

This paper deals with the dynamics of a ball rolling without slipping on the surface of a vertical cylinder whose symmetry axis executes circular motions. The equations of motion of the system considered are integrated by quadratures, and the possibility of the ball moving downward or upward along the cylinder's axis is examined in detail. It turns out that, by making the supporting cylinder execute circular motions, it is impossible to achieve an unbounded elevation of the ball along the cylinder. In the general case, the ball will execute quasi-periodic motions without drifting in the vertical direction either upward or downward. However, there exist resonant initial conditions under which the ball will on average move downward.

Further work in this direction involves the search for a model that could describe the case where the ball moves on average upward due to periodic motions of the supporting surface. In particular, it would be interesting to estimate the influence of friction on the motion of the system considered. It is interesting that, in the problems of a ball rolling on a rotating plane or cone, the trajectories of the ball within the framework of the nonholonomic model are bounded. And adding even small friction leads to a drift of the ball to infinity [20, 21].

Also, it is interesting to consider the oscillations of a cylinder of a more general form, including motions in the vertical direction. In addition, it would be interesting to consider the case where the supporting surface is a circular cone.

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CONFLICT OF INTEREST

The authors declare that they have no conflicts of interest.

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